REVIEW 1 REPORT

Predictive Memory Leak Detection Using Time Series Analysis

Introduction

Memory leaks occur when applications allocate memory but fail to release it after use, causing memory consumption to increase gradually. This degrades system performance, increases resource usage, and may ultimately lead to crashes. Existing techniques mainly detect memory leaks after abnormal behavior has already begun. Therefore, an early prediction mechanism is needed to prevent failures before they occur.

Literature Survey

A survey of research papers from Google Scholar, IEEE, and Springer was conducted. Existing work focuses on static analysis, runtime profiling, heap monitoring, machine learning based detection, and cloud memory monitoring. Most studies successfully identify memory leaks, but they primarily operate after memory degradation becomes visible.

Research Gap

After reviewing multiple research papers, it was observed that most existing approaches are reactive rather than predictive. They mainly rely on memory utilization or runtime analysis and often consider limited system information. Very few approaches combine multiple operating system metrics (RAM usage, CPU utilization, page faults, swap memory, disk I/O and related metrics) with time-series forecasting to predict memory leaks before they become critical.

Problem Statement

Current memory leak detection methods identify leaks only after abnormal memory behavior is observed. This delay may already affect application performance and system stability. There is a need for a lightweight predictive framework capable of forecasting memory leaks early enough to enable preventive action.

Proposed Solution

The proposed system continuously monitors multiple operating system metrics and treats them as time-series data. A forecasting model such as LSTM will learn temporal patterns from historical system behavior and estimate the likelihood of an upcoming memory leak. When abnormal trends are predicted, the system generates an early warning so corrective action can be taken before performance degradation or crashes occur.

Novelty

Unlike conventional approaches that focus mainly on detection, the proposed work emphasizes prediction. By combining multiple operating system metrics with time-series analysis, the system aims to provide proactive memory leak prediction instead of reactive identification.

Objectives

1. Study existing memory leak detection techniques.
2. Collect and analyze multiple operating system metrics.
3. Apply time-series analysis for early prediction.
4. Develop a predictive machine learning model.
5. Evaluate prediction accuracy and system overhead.

Methodology

- Conduct literature survey.
- Acquire publicly available datasets and/or operating system telemetry.
- Preprocess and organize time-series data.
- Train and evaluate forecasting models (e.g., LSTM).
- Compare predicted results with actual memory leak events using suitable evaluation metrics.

Expected Outcome

The proposed system is expected to predict memory leaks before they become critical, reduce unexpected crashes, improve system reliability, and support proactive resource management with minimal runtime overhead.

Conclusion

The literature survey indicates that existing memory leak detection techniques predominantly detect problems after abnormal behavior begins. The proposed predictive framework addresses this gap by

using multiple operating system metrics and time-series forecasting to provide early warning of potential memory leaks.

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